
August 4, 2004

MAC-1G_AMBA Version 2.02E00

Release Notes

Document revision history

Revision	Date	Person	Changes made from previous revision
Rev. 1.00	Aug. 4, 04	T.K.	First revision.

Introduction

The purpose of the Release Notes is to provide summary information regarding new version of IP component. The Release Notes provide fundamental information about the IP component being released and briefly describes enhancements, new features, improvements, etc.

Release notes

Component Information

Name	MAC-1G_AMBA
Version Number	202E00
Previous Version	200E00
Release Date	-
Release Type	Major

Product information

Overview	Bug fix and new features added	
Deliverables	Changed	Description
Documentation	Yes	Description of the new modules added, pinout and block diagram changed.
RTL code	Yes	New files added, modified.
Test Bench	Yes	Modifications according to pinout changes of the core.
Tests	Yes	New tests added, compare results changed.
Scripts and Macros	Yes	Modifications in the synthesis/simulation scripts according to new files added.

New Features

Name	Description
F202.00.fd	Flow Control module added. The flow control receiver module analyzes the incoming stream and captures the Ethernet control frames. When the proper pause frame is detected the module delays a transmission of the next frames for a requested period of time. The flow control transmitter generates pause frames that are inserted into outgoing packet stream to instruct the party host to pause (or re-establish) a transmission. The pause frames are generated according to the actual receive FIFO usage level and a programmable threshold levels.
	New RTL files added to the core: fcr.vhd, fct.vhd, bp.vhd, dom.vhd, domreq.vhd, domack.vhd Statistical counters module added. This module relieves the host and software drivers of the responsibility of maintaining statistics for managed Ethernet port.
F202.00.sc	New RTL files added to the core: sc.vhd, fcstat.vhd. Two DP RAM modules added at the chip level: scdram.vhd.

Enhancement and Improvements

Name	Description
F202.00.ifg	Minimum transmit IFG reduced. Modified RTL files: utility_mac_1g.vhd, tc.vhd
F202.00.cs	CS - Collision Seen functionality in the receive descriptor fixed. Modified RTL files: bd.vhd, rc.vhd, rfifo.vhd, rlsm.vhd.

Bug Fixes

Name	Description
F202.00.gcnt	GCNT - General Purpose Timer reset value fixed. Modified RTL files: csr.vhd
F202.00.db	fixed receive DATA RAM read/write address pointers when receiving a frame with dribbling bit. Modified RTL files: rc.vhd, rfifo.vhd
F202.00.es	ES - receive error summary fixed. Modified RTL files: rlsm.vhd.
F202.00.backoff	fixed backoff algorithm. Modified RTL files: tc.vhd, bd.vhd.
F202.00.fli	fli cross clock domain signal is now properly registered. Modified RTL files: rfifo.vhd.

Open Issues and Known Bugs

Name	Description
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